

## 6.8 Leverage as geometry

The projection  $\mathbf{M}$  is an  $n \times n$  matrix whose entries have a direct interpretation. Since  $\hat{y}_i = \sum_j m_{ij} y_j$ , the entry  $m_{ij}$  is the weight observation  $j$  receives in the fitted value for observation  $i$ . So

$$m_{ij} = \frac{\partial \hat{y}_i}{\partial y_j}.$$

The diagonal entries measure how strongly each observation pulls the fitted surface towards itself.

### Definition 6.8.1 — Leverage

The **leverage** of observation  $i$  is  $h_{ii} = m_{ii}$ , the  $i$ th diagonal entry of  $\mathbf{M}$ . In this context  $\mathbf{M}$  is often called the **hat matrix** and written  $\mathbf{H}$ , because it “puts the hat on”  $\mathbf{y}$ .

Leverage is a property of the design, not of the response. It depends on  $\mathbf{y}$  not at all, and on  $\mathbf{X}$  only through  $\mathcal{C}(\mathbf{X})$ , so it is unchanged by reparameterization.

### Proposition 6.8.1 — Properties of leverage

Let  $\text{rank}(\mathbf{X}) = r$ , and let  $\mathbf{e}_i$  denote the  $i$ th standard basis vector. Then:

- (a)  $h_{ii} = \|\mathbf{M}\mathbf{e}_i\|^2 = \sum_j m_{ij}^2$ , the squared length of the projection of  $\mathbf{e}_i$  onto  $\mathcal{C}(\mathbf{X})$ ;
- (b)  $0 \leq h_{ii} \leq 1$ , and  $\sum_i h_{ii} = r$ , so the average leverage is  $r/n$ ;
- (c) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $h_{ii} \geq 1/n$ ;
- (d)  $h_{ii} = 1$  iff  $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$ . In that case  $\hat{y}_i = y_i$  for every  $\mathbf{y}$ , the  $i$ th residual is always zero, and  $m_{ij} = 0$  for  $j \neq i$ ;
- (e) if  $\mathbf{X} = \mathbf{Q}\mathbf{R}$  with  $\mathbf{Q}$  having orthonormal columns spanning  $\mathcal{C}(\mathbf{X})$ , then  $h_{ii}$  is the squared length of the  $i$ th row of  $\mathbf{Q}$ .

*Proof.*

- (a)  $h_{ii} = \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{M}^\top \mathbf{M} \mathbf{e}_i$ , and  $\mathbf{M} \mathbf{e}_i$  is the  $i$ th column of  $\mathbf{M}$ .
- (b) By Proposition 6.3.4(d),  $0 \leq \|\mathbf{M} \mathbf{e}_i\|^2 \leq \|\mathbf{e}_i\|^2 = 1$ . The sum is  $\text{tr}(\mathbf{M}) = r$ .
- (c) With  $\mathbf{M}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$ , the difference  $\mathbf{M} - \mathbf{M}_0$  is a projection (Theorem 6.5.1), hence nonnegative definite, so  $h_{ii} \geq (\mathbf{M}_0)_{ii} = 1/n$ .
- (d) Equality holds in  $\|\mathbf{M} \mathbf{e}_i\| \leq \|\mathbf{e}_i\|$  iff  $(\mathbf{I} - \mathbf{M})\mathbf{e}_i = \mathbf{0}$ , that is, iff  $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$ . Then  $\mathbf{M} \mathbf{e}_i = \mathbf{e}_i$ , so column  $i$  of  $\mathbf{M}$  is  $\mathbf{e}_i$ , and by symmetry so is row  $i$ . Hence  $\hat{y}_i = y_i$ .
- (e)  $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$ , so  $m_{ii} = \sum_k q_{ik}^2$ .

■

Part (d) describes the extreme case. An observation with leverage one has a coordinate direction of its own in the model space. This happens, for example, when it is the only member of a group in a one-way layout, or when a column of  $\mathbf{X}$  is an indicator for that single observation. The model then fits the observation exactly, whatever its value, and the data give no check on it. High leverage is a weaker form of the same thing. If  $h_{ii}$  is close to one, then  $\hat{y}_i \approx y_i$ , and a gross

error in  $y_i$  barely shows up in its residual.

This shows up in the residual variances. If  $\text{Cov}(\epsilon) = \sigma^2 \mathbf{I}$ , then by Proposition 6.4.3(e)

$$\text{Var}(\hat{\epsilon}_i) = \sigma^2(1 - h_{ii}),$$

so residuals at high-leverage points are *less* variable than others, even though those are exactly the points where an error would do the most damage. [Chapter 20](#) standardizes residuals to correct for this (Definition 20.1.1).

## 6.8.1 Leverage as distance from the centre

When the model has an intercept, leverage has a second description in variable space: distance from the centroid of the regressors. Write  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ , let  $\bar{\mathbf{x}}$  be the vector of column means of  $\mathbf{X}_1$ , and let  $\tilde{\mathbf{X}}_1 = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}_1$  be the centred regressors, assumed to have full column rank.

### Proposition 6.8.2

Under these assumptions,

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where  $\mathbf{x}_{(i)}$  is the  $i$ th row of  $\mathbf{X}_1$ , written as a column vector.

*Proof.* By Lemma 6.6.1 with  $\mathbf{X}_2 = \mathbf{1}$ ,  $\mathbf{M} = n^{-1}\mathbf{1}\mathbf{1}^\top + \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1^\top$ . The  $i$ th row of  $\tilde{\mathbf{X}}_1$  is  $(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top$ . ■

The quadratic form is  $(n-1)^{-1}$  times the squared Mahalanobis distance of  $\mathbf{x}_{(i)}$  from the centroid, measured in the metric of the sample covariance matrix of the regressors. Leverage is therefore high for points that are unusual *jointly*, even when no single coordinate is extreme. A state with moderate poverty and moderate single parenthood, but an unusual *combination* of the two, can have high leverage. For simple regression the formula reduces to  $h_{ii} = n^{-1} + (x_i - \bar{x})^2/S_{xx}$ .

### Example. (High-leverage states)

Return to the state data, now with all 51 jurisdictions including the District of Columbia, and the model of Example (Poverty and murder rates). The average leverage is  $p/n = 0.078$ , and the smallest is 0.021, just above the bound  $1/n$ . The three largest are

| Jurisdiction         | $h_{ii}$ | multiple of average |
|----------------------|----------|---------------------|
| District of Columbia | 0.506    | 6.5                 |
| Mississippi          | 0.212    | 2.7                 |
| Alaska               | 0.177    | 2.3                 |

The District of Columbia is entirely urban, with by far the highest proportion of single-parent households. It sits at a corner of the regressor cloud that no state comes near, and its leverage is about half the maximum possible. It was omitted from Example (Poverty and murder rates). Exercise (A1) asks what happens to the poverty coefficient when it is put back.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data           # District of Columbia
               kept here
X = np.column_stack([np.ones(len(data)), data["poverty"], data["single"], data["
    urban"]])

Q, _ = np.linalg.qr(X)                                # orthonormal basis for C(X)
h = np.sum(Q ** 2, axis=1)                             # hii = mii = ||row i of Q||2
n, p = X.shape
print("sum of leverages:", h.sum(), " (= p =", p, ")")
print("range:", h.min(), h.max(), " bounds:", 1 / n, 1)
top = np.argsort(h)[::-1][:3]
for i in top:
    print(f"{data.index[i]:22s} h = {h[i]:.3f}    ({h[i] / (p / n):.1f} x average)")
```

[Run this code online](#)

The listing computes leverages from a QR factorization rather than by forming  $\mathbf{M}$ . For large  $n$  the  $n \times n$  matrix  $\mathbf{M}$  should never be formed. [Section 6.10](#) explains why.

## 6.8.2 Exercises

### 6.8.2.1 A. Check your understanding

#### Exercise A1.

For simple regression, derive  $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$  directly from Example (Intercept inside simple regression), and show that the largest leverage belongs to the observation with  $x_i$  farthest from  $\bar{x}$ .

### 6.8.2.2 B. Practice

#### Exercise B1.

Show that  $m_{ij}^2 \leq m_{ii}(1 - m_{ii})$  for  $i \neq j$ , and hence  $|m_{ij}| \leq 1/2$ . When is the bound attained?

*Solution.* From [Proposition 6.8.1\(a\)](#),  $m_{ii} = \sum_k m_{ik}^2 \geq m_{ii}^2 + m_{ij}^2$ . So  $m_{ij}^2 \leq m_{ii}(1 - m_{ii}) \leq 1/4$ . Equality needs  $m_{ii} = 1/2$  and  $m_{ij}$  the only other nonzero entry of row  $i$ . The intercept-only model with  $n = 2$ , where every entry of  $\mathbf{M}$  is  $1/2$ , attains it.

**Exercise B2.**

Show that adding a column to  $\mathbf{X}$  cannot decrease any leverage. Show that the leverages in a one-way layout are  $1/n_k$  for members of group  $k$ , and explain why an observation forming a group on its own has leverage one.